

Operator Card — Stage Summary (P21–P29 → P30)

Window: ($\sqrt{2}$ [2.74, 2.92])

Targets: 0.90, 0.94, 0.96, 0.98

Controller: Rate-limited P + deadband; soft integrator (per-target).

Base: (R=5.0), deadband_frac (=0.35), slip_frac (=0.95).

Integrator (default): (=0.20, =0.003, leak=0.9995).

Target 0.98: integrator **off**, deadband_frac (=0.40), slip_frac (=0.97).

KPIs (P29_final):

- 0.94 → median effort **1.16**, chatter **15.5**
- 0.98 → median effort **0.384**, chatter **3**
- Overall % no-slip: **1.0%** (matches P28 baseline)

Ops notes: avoid spike bins flagged in P23; log slip; rate-limit changes.